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1.

Chloroplast is the powerhouse of the cell.

1.

Mitochondria is the powerhouse of the cell.

2.

(B) Carbon dioxide

3.

Photosynthesis is the process by which green plants make their own food using sunlight, water, and carbon dioxide, releasing oxygen.

4.

Plants absorb carbon dioxide through their leaves and water through their roots. Using sunlight captured by chlorophyll, these are converted into glucose and oxygen: $6\text{CO}_2 + 6\text{H}_2\text{O} \rightarrow \text{C}_6\text{H}_{12}\text{O}_6 + 6\text{O}_2$.

5(a).

Chlorophyll absorbs sunlight, mainly in the red and blue regions, providing the energy needed to drive photosynthesis.

5(b).

Light intensity and the concentration of carbon dioxide both affect the rate of photosynthesis.

[rough work] $6\text{CO}_2 + 6\text{H}_2\text{O} \rightarrow ?$ (check coefficients)